CODETA · INTERVIEW SUMMARY

Your code. Better understood.

Armstrong number — scripted QA sample · 11/1/2026

Automatic multiple-choice results and written-answer self-assessments are reported separately. These are informal study records, not official grades. Your code was parsed locally, never compiled or executed.

Automatically graded

— No MC answers

0 correct / 0 recorded multiple-choice questions

Exact answer-key comparison. Skipped MC questions count as incorrect. Written answers never contribute to this score.

Written self-assessment

4	1	0	0
Understood	Partially	Not yet	Unassessed

Chosen by you after reviewing the reference. This is not an objective correctness grade.

Concepts practiced

5 answered / 5 main questions · 0 skipped

cin · if · initialization · function · while

Saved question-order seed: codeta-interface-review

Your next study session

Review **cin**. Explain the cited statement, check its assumptions, and compare with the reference checklist.

Weak topics come from incorrect/skipped MC answers or your written/follow-up self-assessments.

Practice signals

0 hints recorded · 0 skipped

Most practiced: cin (1), if (1), initialization (1), function (1).

Three recent strong, hint-free responses: consider Strict TA next time. Your selected difficulty has not changed.

Question by question

QUESTION 1 · CIN · NORMAL TA · YOUR SOURCE

Self-assessed: partial

What boundary tests would you choose for the input expression at line 5?

Source lines 5–5

```
    if (!(cin >> number) || number < 100 || number > 999) return 1;
```

Your response

Sample explanation for interface review. I would identify the actual condition and explain each relevant path.

Reference explanation

Include a valid value, malformed input, end-of-input, and values outside the expected domain or representable range as relevant. Explain the actual stream-state and validation paths.

Expected points to check

- Test good input, bad input, missing input, and the edges of the allowed range.
- Identify the cited evidence and state the assumptions your argument needs.

QUESTION 2 · IF · NORMAL TA · YOUR SOURCE

Self-assessed: complete

For the if beginning at line 13, propose inputs that exercise its true and false paths. What must your explanation include?

Source lines 13–14

```
    if (sum == number) cout << "Armstrong";
    else cout << "Not Armstrong";
```

Your response

Scripted sample response for layout verification, not an automatically graded student answer.

Reference explanation

Use the exact condition and earlier constraints, justify whether each proposed input can reach this test, and describe the selected branch. Do not assume every syntactic path is reachable.

Expected points to check

- Give a reason for each condition result and check that earlier code lets the input reach it.
- Identify the cited evidence and state the assumptions your argument needs.

QUESTION 3 · INITIALIZATION · NORMAL TA · YOUR SOURCE

Self-assessed: complete

The source initializes temp from another named value. Why might keeping two independent values be useful?

Source lines 6–6

```
    int temp = number;
```

Your response

Scripted sample response for layout verification, not an automatically graded student answer.

Reference explanation

A value copy can preserve an earlier value while another object is modified. Explain whether later uses in this actual program support that role; the initializer alone does not reveal intent.

Expected points to check

A copy can keep an old value. Look at later uses before calling it a backup.

Identify the cited evidence and state the assumptions your argument needs.

QUESTION 4 · FUNCTION · NORMAL TA · YOUR SOURCE

Self-assessed: complete

Present main to your TA by separating its input contract, work, and result. What should you avoid assuming?

Source lines 3–16

```
int main() {
    int number;
    if (!(cin >> number) || number < 100 || number > 999) return 1;
    int temp = number;
    int sum = 0;
    while (temp > 0) {
        int digit = temp % 10;
        sum += digit * digit * digit;
        temp /= 10;
    }
    if (sum == number) cout << "Armstrong";
    else cout << "Not Armstrong";
    return 0;
}
```

Your response

Scripted sample response for layout verification, not an automatically graded student answer.

Reference explanation

Use actual parameters, body, and return statements; state valid inputs and side effects. Do not infer a correct contract or behavior solely from the function’s name.

Expected points to check

- Explain what comes in, what happens, and what comes back using the actual source.
- Identify the cited evidence and state the assumptions your argument needs.

QUESTION 5 · WHILE · NORMAL TA · YOUR SOURCE

Self-assessed: complete

Present the loop section on lines beginning at 8 to your TA: explain its condition, body updates, and possible exits.

Source lines 8–12

```
while (temp > 0) {
    int digit = temp % 10;
    sum += digit * digit * digit;
    temp /= 10;
}
```

Your response

Scripted sample response for layout verification, not an automatically graded student answer.

Reference explanation

Name the actual condition and changes in the cited source, describe when it is tested, and identify any exits. Justify the intended behavior rather than inferring purpose from variable names.

Expected points to check

- Say what permits entry, what changes, and what could end the loop.
- Identify the cited evidence and state the assumptions your argument needs.